METABOLIC STABILITY ASSAY IN SUSPENSION CRYOPRESERVED HEPATOCYTES

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SIGNATURES

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
%	Percentage
°C	Degrees Celsius
µL	Microlitre
µM	Micromolar
µm	Micrometre
COA	Certificate of analysis
DMPK	Drug metabolism and pharmacokinetics
DMSO	Dimethyl sulfoxide
g	Gram
HCl	Hydrochloric acid
HOD	Head of the department
HPLC	High performance liquid chromatography
hr	Hour
IPM	Intellectual property management
LNB	Lab note book
M	Molarity
mAU	Milli absorbance unit
mg	Milligram
min	Minutes
mL	Millilitre
mM	Millimolar
MS	Microsoft
N	Normality
NaOH	Sodium hydroxide
OPA	Ortho phosphoric acid
PD	Pharmaceutical development
pH	Potential of hydrogen
PPE	Personal protective equipment
rpm	Revolutions per minute
S. No.	Serial number

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1. OBJECTIVE

The objective of this experiment is to study the metabolic stability of test compound(s) in cryopreserved rat and human hepatocytes (metabolism qualified suspension cells).

2. SCOPE

The procedure described in this protocol will be used for routine testing of metabolic stability of test compound(s) in cryopreserved hepatocytes (human and rat).

3. RESPONSIBILITY

All personnel performing this study are responsible to follow the process outlined in this protocol, at Aurigene Pharmaceutical Services Ltd, India.

4. INTRODUCTION

Liver is the primary site of metabolism for most of the xenobiotics including drugs. Liver-derived systems such as liver slices, sub-cellular liver fractions and intact hepatocytes are typically utilized to assess the metabolism of compounds. Intact hepatocytes contain full complement of hepatic drug metabolizing enzymes the cytochrome P450s, other non-P450 enzymes and phase II enzymes such as sulfo and glucuronosyltransferase. Hepatocytes represent a prime model system for studying drug disposition in vitro. Compounds can be screened and rank-ordered according to metabolic half-life estimates or in vitro intrinsic clearance values obtained from metabolic stability studies. The use of cryo-preserved or freshly isolated hepatocytes from different species enables an understanding of interspecies differences in metabolism and hepatic clearance of the compound.

Cryopreserved hepatocytes retain enzymatic activities similar to those of fresh hepatocytes and offer convenience to the end user. The utility of cryopreserved hepatocytes to define in vitro drug disposition is advantageous as compared to other model systems.

5. SAFETY AND PRECAUTIONS

During the conduct of the study, the study personnel should take necessary safety precautions and use personal protective equipment (PPE) such as facemasks, gloves, lab coat, goggles, shoes etc.

DMSO:

- Keep away from sources of ignition.
- Do not breathe vapour or mist.
- Avoid contact with skin, eyes and clothing.

6. MATERIALS AND EQUIPMENT

6.1 EQUIPMENT

Equipment	Make
96-well deep well plates (1.2 mL)	Axygen Scientific
96-well deep well plates-V bottom (1.2 mL)	Phenomenex
96-well flat bottom TC plates	Corning
Analytical balance	Denver/ Sartorius
Centrifuge with plate/ tube rotors	Eppendorf/ Kubota & Thermo

Equipment	Make
Deep freezer (-20°C)	Thermo Scientific/ Celfrost
Haemocytometer	Vasa scientific co.
Incubator	Heidolph incubator 1000 with Titramax 1000
Inverted microscope	Nikon
LC-MS/MS (API-3200, API-5500)	MDS/ AB Sciex
Liquid nitrogen tank	Thermo Scientific
Mixmate	Eppendorf
Refrigerator	LG
Software	Graph pad prism/ Microsoft excel
Water bath	Julabo corporation

Note:

- Equivalent equipment/ instruments could be used based on the availability.

6.2 CHEMICALS

Chemicals	Make; Catalogue No.
Acetonitrile	J.T. Baker; 9012-03
Buspirone	Sigma; B7148
Carbazeran	Sigma; SML0308
Dimethyl sulfoxide (DMSO)	Sigma Aldrich; D8418
Ethanol	Changshuhong sheng fine chemic¬¬als Co, ltd; 20160503
Glyburide	Sigma Aldrich; G2539
Hepatocyte Thaw Medium	Bioreclamation IVT; Z99019
Leibovitz's L-15 Medium	Life technologies; 21083027
Midazolam	RA ChemPharma Limited; ADM/04/003/18
OH-Coumarin (Umbelliferone)	Aldrich; H24003
Ramipril	Sigma; R0404
Trypan blue	Sigma; T8154

Note:

- Complete list of chemicals used and their details should be provided in the LNB.
- Equivalent grade chemicals/ reagents from alternate sources can be used on need basis.
- Human and rat cryopreserved hepatocytes (metabolism qualified) can be procured from certified vendors with COA.
- Source information of hepatocytes should be provided along with results.

6.3 PREPARATION OF SOLUTIONS AND STORAGE CONDITIONS

S. No.	Composition	Storage
1.	Thaw medium HT medium	2-8°C
2.	Incubation medium Leibovitz's L-15 Medium	2-8°C
3.	Preparation of stop solution Prepare 1 mg/mL DMSO stock solutions of the internal standard (Glyburide) and dilute with acetonitrile to obtain appropriate concentration.	2-8°C
4.	Preparation of test compound(s) Prepare 1 mM intermediate stock solution in DMSO from the higher stock. Dilute the stock solutions in L-15 medium to obtain 2 µM working concentration maintaining DMSO ≤ 0.1%.	Working solution: 20-25°C Prepare fresh
5.	Preparation of reference compound(s) Prepare 4 mM main stock solutions of Midazolam, Ramipril, Carbazeran and 40 mM of OH-Coumarin. Dilute the stock solutions in L-15 medium to obtain 2 and 20 µM working concentration. (For example: Aliquot 1 µL each from main stock solutions in 2000 µL L-15 medium to obtain the cocktail working concentration of reference standards.	Stock solution: -20°C Working solution: 20-25°C Prepare fresh

Note:

- Changes to the above mentioned solution preparation methods should be captured in the LNB & report (If any).
- Final concentration in reaction; 1) Cell density = 1 million/mL 2) Test concentration = 1 µM.

7. PROCEDURE

7.1 THAWING CRYOPRESERVED HEPATOCYTES

Incubate the hepatocyte thaw medium and incubation medium at 37°C on a water bath for 30 min.

Remove cryopreserved hepatocytes vial(s) from the liquid nitrogen storage tank and loose the cap of the vial to ease out the pressure and re-tighten it again

Thaw the vials on water bath at 37°C until the solid mass leave the vial wall and become half frozen.

Once vials are thawed, spray 70% ethanol and transfer the thaw media and vial to a biosafety cabinet. Open the vials and transfer the contents into a 45mL falcon tube containing the thaw medium and rinse it with thaw medium to remove left out cells

Centrifuge the tube at 100 x g for 10 min at 25°C.

Discard the supernatant. Add ~2 mL of incubation medium and mix gently to suspend the cells uniformly.

Dilute a small aliquot and count the viability of cells by trypan blue exclusion method using a haemocytometer. The viability of the cells should be >70%.

Dilute cells with incubation medium to obtain a working cell density of 2.0 x 10^6 cells/mL.

7.2 SCREENING OF TEST COMPOUND(S) at 1 µM CONCENTRATION

Assay is performed in singlet (n=1). Buspirone and cocktail (Carbazeran, Ramipril, 7-OH-Coumarin and Midazolam) should be included as reference standards in each assay.

Dispense 100 µL working stock of hepatocyte cell suspension in required wells of 96 flat bottom assay plate.

Pre Incubate the assay plate at 37°C for 15 min at 600 rpm.

Start the reaction by adding 100 µL of working solution of test/ reference compounds and mix well.

Draw 25 µL incubation samples from assay plate and dispense into T_1 wells of 96 deep well plates containing 100 µL ice cold stop solution (with 250 ng/mL IS). Store the plate at 2-8°C till the completion of the assay.

Incubate the reaction plate further at 37°C at 600 rpm for 80 min.

At each time point (T_{10}, T_{20}, T_{40}, T_{60} and T_{80} min), draw 25 µL samples and transfer to designated wells in the 96 deep well plate containing 100 µL ice cold stopping solution.

Centrifuge the samples at 4000 rpm for 10 min at 4°C.

Withdraw 90 µL of supernatant and add to 100 µL of water.

Submit the supernatant for analysis by LC-MS/MS.

If not analysed immediately, store the samples at 4°C

8. ASSAY CONTROL

Midazolam, Carbazeran, Ramipril and 7-OH-Coumarin should be included as assay control.

Study would be qualified based on the acceptance criteria given in the data template.

Note:

- Any changes to the assay controls would be made only with the consent of the sponsor and should be described in the LNB & report ((If any).

9. DATA RECORDING

The data should be tabulated in MS excel based metabolic stability assay (hepatocyte) template and the final data should be pasted in the LNB. Raw data should be recorded in the LC-MS/MS server with the study number and the data path should be provided in the excel file and LNB.

10. DATA ANALYSIS

Metabolic stability is defined as the amount of substrate metabolized upon the incubation with hepatocytes. Metabolic stability is assessed from the percent parent compound remaining at various time points based on the change in analyte to internal standard peak-area ratio with respect to maximal peak area ratio.

Calculate following parameters using the given formula:

a) % parent remaining at X time:

$$\% \text{ parent remaining} = \frac{\text{Peak area ratio at X time point}}{T_{max} \text{ peak area ratio}} \times 100$$

b) Half-life (min):

Graph pad prism used to calculate the t½.

c) CL_{int} (uL min^{-1} million $cells^{-1}$):

$$CL_{int} = \frac{\text{Incubation volume}}{\text{No. of cells in incubation}} \times (0.693/\text{Half-life})$$

11. REPORTING

Data should be reported in an MS excel format. Based on the need, MS word based reports may also be prepared.

If reference compound(s) does not meet the specified acceptance criteria, the complete batch should be rejected.

Condition	**Report Status**	**Remark**
Precipitation observed when preparing compound plate	Undefined	Precipitation in DMSO stock solution
Test compound not detectable	Undefined	Analytical issue - No peak detected
Double peak observed in chromatogram for assay and also appears during optimization check. Clint > factor of 2 depending upon integration chosen.	No data reported	Analytical issue - Double peak in chromatogram
Unidentified double peak observed in chromatogram for assay but not present during optimization check, question over data validity.	No data reported	Analytical issue - Double peak in chromatogram
Poor peak shape observed and accuracy of result compromised	No data reported	Analytical issue- Poor chromatogram
Test compound not detected by T_{10} min due to high clearance. T_1 min peak area high (Agrees with optimization)	Report Clint > 250 & $t_{1/2}$ < 2.77	N/A
Test compound not detected by T_{10} min due to high clearance. T_1 min peak area low (Disagrees with optimization)	No data reported	Very high clearance likely
Clint calculated to be greater than 250	Report Clint > 250 & $t_{1/2}$ < 2.77	N/A
Clint calculated to be less than 4 (all data points should be within 60 & 100%)	Report Clint < 4 & $t_{1/2}$ > 175	N/A
Clint calculated ≤ 9 (SE % fit should be < 50% even also one value was	Report Clint value	N/A

Condition	Report Status	Remark
outliered from 6 time points	calculated	
Clint calculated to be >9 (SE % fit should be< 25% even also one value was outliered from 6 time points)	Report Clint value calculated	N/A

12. DATA ARCHIVAL

All the raw data, chromatograms and reports should be stored in the server at Aurigene Pharmaceutical Services Limited, India for a period of five years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and the same should be archived as per the instructions/ procedure of Intellectual Property Management (IPM) department.

13. REFERENCES

- Kola I, Landis J Nat Rev Drug Discovery 3: 711–715; 2004.
- US FDA draft guidance for Industry: Drug interaction studies—study design, Data analysis, and Implications for dosing and labelling, US Food and drug administration publication; 2012

14. CHANGE AND AMENDMENT HISTORY

Change History		
Version	**Changes made/ Reason for changes**	**Effective date**
2.0	Revised Version	01 Jul 2022